

TRACE

Tracking, Recognition, Analytics & City-wide Traffic Enforcement

Team Roles & Milestone Contribution Guide — Revised v1.1

Current execution team: 3 members. The six milestones and all TRACE requirements remain unchanged.

1. Role Assignment

Role	Focus
P1 — Frontend Developer / UI Engineer	Designs and codes the complete React/MapLibre UI and integrates REST/WebSocket data.
P2 — Backend + ML Engineer	Builds the complete Python/FastAPI pipeline, PostgreSQL/PostGIS layer and all five logical intelligence modules plus alerts.
P3 — Data Collector & Project Support	Prepares datasets, annotations, ground truth and demo scenarios; supports QA, documentation and rehearsal.

2. P1 — Frontend Developer / UI Engineer

M1

- React/Vite shell, navigation and all 8 routes.
- Reusable UI components and visual system.

M2

- Confidence Badge, Status Dot, Map Panel and state components.

M3

- Evidence Panel and identity-confidence presentation.

M4

- Vehicle Trace, GIS trajectory, observation timeline and anomaly treatment.

M5

- Dashboard, Heatmap, OD Matrix and Segment Detail with real API integration.

M6

- Alerts, Blacklist, Camera Network Status and WebSocket live-update UI; final polish.

3. P2 — Backend + ML Engineer

All backend intelligence remains logically separated but runs in one FastAPI application.

M1

- PostgreSQL + PostGIS and existing schema.
- Single FastAPI app with internal perception, identity, reasoning, analytics, prediction and alerts modules.
- Seed cameras/road_edges and configure Docker Compose.

M2

- YOLO + ByteTrack + PaddleOCR + temporal OCR fusion.
- Persist ocr_reads and vehicle_observations.

M3

- Identity Score, low-confidence fallback, identity_matches and camera-reliability weighting.

M4

- Reachability, trajectory reconstruction, impossible-journey detection and existing trajectory endpoint.

M5

- Density, OD, congestion, forecast and existing analytics endpoints.

M6

- Blacklist, anomaly alerts, alert logging, access control and existing WebSocket endpoint.

4. P3 — Data Collector & Project Support

M1

- Collect/organize footage; label ground truth; prepare evaluation and demo data.

M2

- Maintain annotation quality and support condition-based OCR evaluation.

M3

- Prepare known observation pairs and validate identity outcomes.

M4

- Stage known journeys including one impossible pair; verify timestamps and expected outcomes.

M5

- Validate density, OD and congestion results against known demo data.

M6

- Run end-to-end QA, test alerts/camera-down/empty/low-confidence states, support docs and rehearsal.

5. Working Rule

P2 is the critical path in M2–M4. P1 can use mocked JSON/API responses until endpoints are ready. P3 works ahead on data and validation. All three review milestone integrations.

6. Files That Remain Unchanged

The Backend Schema, App/Website Flow and UI/UX Brief remain unchanged. The PRD was separately revised to v1.1 for Section 7.2 team-composition wording only — no functional requirement changed. The TRD, Implementation Plan and this team guide are the only documents revised for the single-FastAPI deployment and three-person ownership.